

Greenhouse Climate Controller

Match plants to the right climate zone

At a glance: Run the greenhouse controls for a plant that cannot complain, only wilt. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: controlled environments, temperature, humidity, light, plant needs, tradeoffs (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- plant profile cards: one set per team (4, plus 1 spare)
- control dial boards: one per team (4, plus 1 spare)
- dry erase markers: shared, 1 to 2 per team: one per team (4, plus 2 spare)
- clipboards: one per team (4)
- dry erase markers: shared, 1 to 2 per team

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Set up one station per team with the materials above, sorted and ready.
- 02 Stage the scoring sheet and any reference values before testing begins.

3 No PPE is needed: the tabletop phase is a dry card-and-board game with no chemicals, soil, or water. For the greenhouse tour, confirm the route and off-limits plants and controls with greenhouse staff, and check the roster for plant or pollen allergies.

- 01 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Run the greenhouse controls for a plant that cannot complain, only wilt." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve

0:18 - 0:44

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Match and set

Teams sort plant cards by need, match each plant to a zone, and set temperature, humidity, and light on the dial board using their tour evidence.

Phase 6 Resolve tradeoffs

Where two plants conflict in a zone, or raising temperature would dry the air too much, each team picks the setting its evidence best supports and notes the tradeoff it accepted.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. Wipe the dry-erase boards and return all cards, boards, and clue sheets to the bin so the next team has a complete set; there is no PPE to remove and nothing to wash.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: A team places a plant with no supporting tour evidence. **Fix:** Ask which clue from the tour backs that zone choice; if they have none, send them back to their clue sheet to find or admit the gap before scoring.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to controlled environments.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which plant was hardest to place, and why?
- What tradeoff did you have to accept?
- How do real growers handle conflicting needs?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Correct setting choices	30
Use of tour evidence	25
Explanation	20
Layout efficiency	15
Teamwork	10
Total	100

CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

SOURCES Temple Ambler Greenhouse